MPLS L3VPN - Full Router Configuration

VRF + IS-IS + MPLS/LDP + MP-BGP + OSPF

Lab routers: PE-CASA - P - PE-BM - CIH CEs - BMCE CEs

1. Addressing Summary

Router	Interface / Role	IP Address	Purpose
PE-CASA	Loopback0	1.1.1.1/32	Router ID / LDP / MP-BGP
PE-CASA	Fa0/0 -> P	10.0.0.2/30	Provider core
PE-CASA	Fa1/0 -> CIH-CASA	172.16.10.1/30	VRF CIH
PE-CASA	Fa1/1 -> BMCE-CASA	172.16.30.1/30	VRF BMCE
P	Loopback0	2.2.2.2/32	Router ID / LDP
P	Fa0/0 -> PE-CASA	10.0.0.1/30	Provider core
P	Fa1/0 -> PE-BM	10.0.1.1/30	Provider core
PE-BM	Loopback0	3.3.3.3/32	Router ID / LDP / MP-BGP
PE-BM	Fa0/0 -> P	10.0.1.2/30	Provider core
PE-BM	Fa1/0 -> CIH-BM	172.16.20.1/30	VRF CIH
PE-BM	Fa1/1 -> BMCE-BM	172.16.40.1/30	VRF BMCE
CE-CIH-CASA	Fa0/0 -> PE	172.16.10.2/30	WAN
CE-CIH-CASA	Fa1/0 -> LAN	192.168.10.1/24	CIH CASA LAN
CE-CIH-BM	Fa0/0 -> PE	172.16.20.2/30	WAN
CE-CIH-BM	Fa1/0 -> LAN	192.168.20.1/24	CIH BM LAN
CE-BMCE-CASA	Fa0/0 -> PE	172.16.30.2/30	WAN
CE-BMCE-CASA	Fa1/0 -> LAN	192.168.30.1/24	BMCE CASA LAN
CE-BMCE-BM	Fa0/0 -> PE	172.16.40.2/30	WAN
CE-BMCE-BM	Fa1/0 -> LAN	192.168.40.1/24	BMCE BM LAN

Note: if a CE LAN interface in your GNS3 appliance uses a different port number, keep the same IP and replace only the interface name.

2. PE-CASA - Full Configuration

PE-CASA hosts both customer VRFs, participates in IS-IS/MPLS toward P, and forms the MP-BGP VPNv4 session with PE-BM.

```
enable
configure terminal
hostname PE-CASA

! ----- VRFs -----
ip vrf CIH
 rd 65000:1
 route-target export 65000:1
 route-target import 65000:1
 exit

ip vrf BMCE
 rd 65000:2
 route-target export 65000:2
 route-target import 65000:2
 exit

! ----- Provider loopback -----
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
 ip router isis CORE
```

```
 no shutdown
 exit

! ----- Core link to P -----
interface FastEthernet0/0
 description TO-P
 ip address 10.0.0.2 255.255.255.252
 ip router isis CORE
 mpls ip
 no shutdown
 exit

! ----- CIH customer link -----
interface FastEthernet1/0
 description TO-CE-CIH-CASA
 ip vrf forwarding CIH
 ip address 172.16.10.1 255.255.255.252
 no shutdown
 exit

! ----- BMCE customer link -----
interface FastEthernet1/1
 description TO-CE-BMCE-CASA
 ip vrf forwarding BMCE
 ip address 172.16.30.1 255.255.255.252
 no shutdown
 exit

! ----- IS-IS provider IGP -----
router isis CORE
 net 49.0001.0000.0000.0001.00
 is-type level-2-only
 exit

! ----- MPLS / LDP -----
mpls label protocol ldp
mpls ldp router-id Loopback0 force

! ----- MP-BGP VPNv4 -----
router bgp 65000
 bgp log-neighbor-changes
 neighbor 3.3.3.3 remote-as 65000
 neighbor 3.3.3.3 update-source Loopback0
 address-family vpnv4
  neighbor 3.3.3.3 activate
  neighbor 3.3.3.3 send-community extended
 exit-address-family

 ! VRF CIH -> MP-BGP
 address-family ipv4 vrf CIH
  redistribute connected
  redistribute ospf 10
 exit-address-family

 ! VRF BMCE -> MP-BGP
 address-family ipv4 vrf BMCE
  redistribute connected
  redistribute ospf 20
 exit-address-family
 exit

! ----- OSPF CE-PE for CIH -----
router ospf 10 vrf CIH
 router-id 10.10.10.10
 network 172.16.10.0 0.0.0.3 area 0
 redistribute bgp 65000 subnets
 exit
```

```
! ----- OSPF CE-PE for BMCE -----
router ospf 20 vrf BMCE
 router-id 20.20.30.10
 network 172.16.30.0 0.0.0.3 area 0
 redistribute bgp 65000 subnets
 exit

end
write memory
```

3. P - Full Configuration

The P router has no customer VRFs and no MP-BGP VPNv4 session. It provides provider-core IP reachability and MPLS label switching.

```
enable
configure terminal
hostname P

interface Loopback0
 ip address 2.2.2.2 255.255.255.255
 ip router isis CORE
 no shutdown
 exit

interface FastEthernet0/0
 description TO-PE-CASA
 ip address 10.0.0.1 255.255.255.252
 ip router isis CORE
 mpls ip
 no shutdown
 exit

interface FastEthernet1/0
 description TO-PE-BM
 ip address 10.0.1.1 255.255.255.252
 ip router isis CORE
 mpls ip
 no shutdown
 exit

router isis CORE
 net 49.0001.0000.0000.0002.00
 is-type level-2-only
 exit

mpls label protocol ldp
mpls ldp router-id Loopback0 force

end
write memory
```

4. PE-BM - Full Configuration

PE-BM mirrors PE-CASA: same VRF names and route-targets, separate local CE-facing subnets, and an MP-BGP VPNv4 session sourced from Loopback0.

```
enable
configure terminal
hostname PE-BM
```

```
! ----- VRFs -----
ip vrf CIH
 rd 65000:1
 route-target export 65000:1
 route-target import 65000:1
 exit

ip vrf BMCE
 rd 65000:2
 route-target export 65000:2
 route-target import 65000:2
 exit

interface Loopback0
 ip address 3.3.3.3 255.255.255.255
 ip router isis CORE
 no shutdown
 exit

interface FastEthernet0/0
 description TO-P
 ip address 10.0.1.2 255.255.255.252
 ip router isis CORE
 mpls ip
 no shutdown
 exit

interface FastEthernet1/0
 description TO-CE-CIH-BM
 ip vrf forwarding CIH
 ip address 172.16.20.1 255.255.255.252
 no shutdown
 exit

interface FastEthernet1/1
 description TO-CE-BMCE-BM
 ip vrf forwarding BMCE
 ip address 172.16.40.1 255.255.255.252
 no shutdown
 exit

router isis CORE
 net 49.0001.0000.0000.0003.00
 is-type level-2-only
 exit

mpls label protocol ldp
mpls ldp router-id Loopback0 force

router bgp 65000
 bgp log-neighbor-changes
 neighbor 1.1.1.1 remote-as 65000
 neighbor 1.1.1.1 update-source Loopback0
 address-family vpnv4
  neighbor 1.1.1.1 activate
  neighbor 1.1.1.1 send-community extended
 exit-address-family

 address-family ipv4 vrf CIH
  redistribute connected
  redistribute ospf 10
 exit-address-family

 address-family ipv4 vrf BMCE
  redistribute connected
  redistribute ospf 20
```

```
 exit-address-family
 exit

router ospf 10 vrf CIH
 router-id 10.10.20.10
 network 172.16.20.0 0.0.0.3 area 0
 redistribute bgp 65000 subnets
 exit

router ospf 20 vrf BMCE
 router-id 20.20.40.10
 network 172.16.40.0 0.0.0.3 area 0
 redistribute bgp 65000 subnets
 exit

end
write memory
```

5. CE-CIH-CASA - Full Configuration

```
enable
configure terminal
hostname CE-CIH-CASA

interface FastEthernet0/0
 description TO-PE-CASA
 ip address 172.16.10.2 255.255.255.252
 no shutdown
 exit

interface FastEthernet1/0
 description CIH-CASA-LAN
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit

router ospf 10
 router-id 10.10.10.1
 network 172.16.10.0 0.0.0.3 area 0
 network 192.168.10.0 0.0.0.255 area 0
 exit

end
write memory
```

6. CE-CIH-BM - Full Configuration

```
enable
configure terminal
hostname CE-CIH-BM

interface FastEthernet0/0
 description TO-PE-BM
 ip address 172.16.20.2 255.255.255.252
 no shutdown
 exit

interface FastEthernet1/0
 description CIH-BM-LAN
 ip address 192.168.20.1 255.255.255.0
 no shutdown
```

```
 exit

router ospf 10
 router-id 10.10.20.1
 network 172.16.20.0 0.0.0.3 area 0
 network 192.168.20.0 0.0.0.255 area 0
 exit

end
write memory
```

7. CE-BMCE-CASA - Full Configuration

```
enable
configure terminal
hostname CE-BMCE-CASA

interface FastEthernet0/0
 description TO-PE-CASA
 ip address 172.16.30.2 255.255.255.252
 no shutdown
 exit

interface FastEthernet1/0
 description BMCE-CASA-LAN
 ip address 192.168.30.1 255.255.255.0
 no shutdown
 exit

router ospf 20
 router-id 20.20.30.1
 network 172.16.30.0 0.0.0.3 area 0
 network 192.168.30.0 0.0.0.255 area 0
 exit

end
write memory
```

8. CE-BMCE-BM - Full Configuration

```
enable
configure terminal
hostname CE-BMCE-BM

interface FastEthernet0/0
 description TO-PE-BM
 ip address 172.16.40.2 255.255.255.252
 no shutdown
 exit

interface FastEthernet1/0
 description BMCE-BM-LAN
 ip address 192.168.40.1 255.255.255.0
 no shutdown
 exit

router ospf 20
 router-id 20.20.40.1
 network 172.16.40.0 0.0.0.3 area 0
 network 192.168.40.0 0.0.0.255 area 0
 exit
```

```
end
write memory
```

9. Verification Commands

```
! Provider IGP
show isis neighbors
show ip route isis

! MPLS / LDP
show mpls ldp neighbor
show mpls forwarding-table
show mpls ldp bindings

! VRF
show ip vrf
show ip route vrf CIH
show ip route vrf BMCE

! MP-BGP
show ip bgp vpnv4 all summary
show ip bgp vpnv4 all

! CE-PE OSPF
show ip ospf neighbor
show ip route

! End-to-end tests (examples)
ping 192.168.20.1
ping 192.168.40.1
```